Electromagnetism

A Treatise on Electricity and Magnetism

IISER Bhopal · Saathvik Bogam

My Website

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Chapter 1

Vector Calculus

> ”Then not Euclid, but elementary vectors, conjoined with algebra, and applied to geometry...”
>
> *Electro–Magnetic Theory, 1893*
> Oliver Heaviside

1.1 Grad, Div and Curl

We talk about importance of the gradient, the divergence and the curl, and how one applied them to scalar and vector fields.

1.1.1 Scalar and Vector Fields

A field is an object that prescribes a different object to every point in space. If the prescribed is a scalar, then it is a scalar field and takes the form of a function of (usually) three variables $\phi(x, y, z)$. If the prescribed object is a vector, then it is a vector field denoted by $\boldsymbol{v}(x, y, z)$. Each component of a vector field is an individual scalar field in the each direction.

$$\boldsymbol{v}(x, y, z) = v_x(x, y, z)\hat{\boldsymbol{x}} + v_y(x, y, z)\hat{\boldsymbol{y}} + v_z(x, y, z)\hat{\boldsymbol{z}} \tag{1.1}$$

1.1.2 Gradient

Definition 1.1 – Nabla Operator

One defines the Nabla/Del operator to be the following:

$$\nabla = \hat{\boldsymbol{x}}\frac{\partial}{\partial x} + \hat{\boldsymbol{y}}\frac{\partial}{\partial y} + \hat{\boldsymbol{z}}\frac{\partial}{\partial z} \tag{1.2}$$

We shall discuss how one applies this to scalar fields $\phi(x, y, z)$. The vector $\nabla\phi(\boldsymbol{r})$ is called the gradient of the scalar field ϕ at a point $\boldsymbol{r} = (x, y, z)$.

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial\phi}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial\phi}{\partial z}\hat{\boldsymbol{z}} \tag{1.3}$$

To interpret what this means, we make use of the multivariate chain rule to write the differential of ϕ as:

$$d\phi = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz \tag{1.4}$$

If we define a displacement vector $d\boldsymbol{l} = dx\hat{\boldsymbol{x}} + dy\hat{\boldsymbol{y}} + dz\hat{\boldsymbol{z}}$, then we can rewrite the differential of ϕ as:

$$d\phi = \nabla\phi \cdot d\boldsymbol{l} = |\nabla\phi||d\boldsymbol{l}|\cos\theta \tag{1.5}$$

If we fix the magnitude of $d\boldsymbol{l}$, then it turns out the maximum change in ϕ happens when the angle between $d\boldsymbol{l}$ and $\nabla\phi$ is zero, i.e. $\cos\theta = 1$. Therefore, we say that *the gradient points in the direction of maximum change, and its magnitude gives us the rate of increase in this maximal direction.*

If the gradient of a scalar field vanishes, then that point is a stationary point which could be a maximum (a summit), a minimum (a valley), a saddle point (a pass), or a “shoulder.

1.1.3 Divergence

The dot product of the nabla operator with a vector field $\boldsymbol{v}$ is called the divergence of that vector field.

$$\nabla \cdot \boldsymbol{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \tag{1.6}$$

The divergence of a vector about a point $\boldsymbol{r}$ gives the value of the ”spread” of the vector field. One way to imagine it is by considering the flow of water. If you drop some water and you see it spread out or diverge, then the divergence there is positive, since there is a spread of water. Using this, we define the terms ”faucet” or ”source” for a point with positive divergence and ”sink” or ”drain” for negative divergence. If a point has zero divergence, there is a equal amount of source and sink there, or no flow at all.

1.1.4 Curl

The cross product operation of nabla on a vector field is called the curl of the vector field,

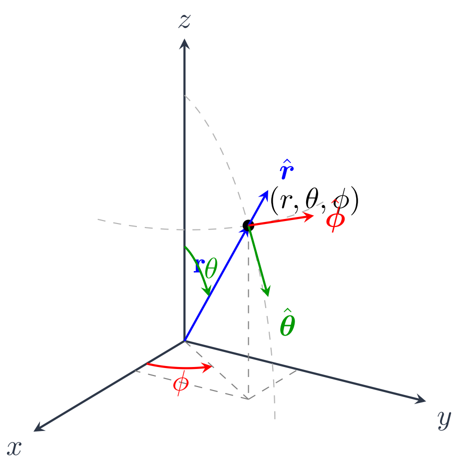
$$\nabla \times \boldsymbol{v} = \sum_{i,j,k} \epsilon_{ijk}\frac{\partial v_j}{\partial r_k} \tag{1.7}$$

where the index go in the order x, y, z and r_k is the k coordinate. It is reccomended to do cross products using the Levi–Civita symbol ϵ_{ijk} (please do this, it made my life doing cross product so much easier), which is defined as:

$$\epsilon_{ijk} = \begin{cases} 1, & \text{cyclic permutation } (xyz, yzx, zxy) \\ -1, & \text{anticyclic permutation } (zyx, yxz, xzy) \\ 0, & \text{otherwise} \end{cases} \tag{1.8}$$

The curl is a measure of how much and what direction the vector $\boldsymbol{v}$ swirls around the point in question. For example, a voxtex or a whirlpool is an example of a large magnitude of curl.

1.2 Curvilinear Coordinates



Let (x, y, z) be the position of a point in cartesian coordinates. Then we define (r, θ, ϕ) to be the position using curvilinear/spherical coordinates. r is the distance of the point from the origin, θ is the angle made with the $+z$–axis, and ϕ is the angle made by the projection on the x–y plane of the line joining the point and origin with the x–axis. Using the given diagram, one can discern that:

$$x = r \sin\theta \cos\phi \tag{1.9}$$

$$y = r \sin\theta \sin\phi \tag{1.10}$$

$$z = r \cos\phi \tag{1.11}$$

And the cartesian to curvilinear transformation being:

$$r = \sqrt{x^2 + y^2 + z^2} \in [0, \infty) \tag{1.12}$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right) \in (0, \pi) \tag{1.13}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) \in (0, 2\pi) \tag{1.14}$$

The unit vectors for the curvilinear coordinates are not fixed unlike the cartesian ones.